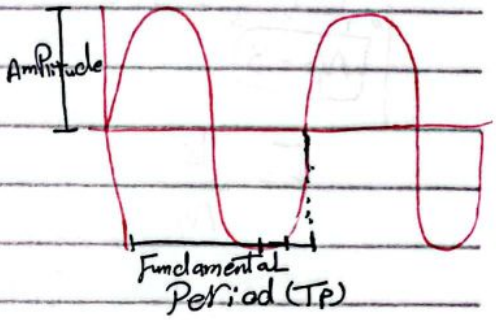
Chapter 1

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#Continous & Discrete Time Signal

* Continous-Time SinuSoidal Signal *

عند أي Time ليها قيمة



$$X(t) = A\sin(2\pi Ft + \theta)$$

Where:

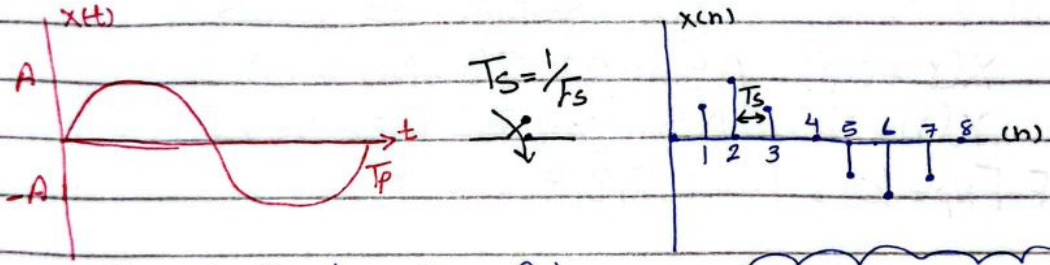
A → Amplitude.

F → Freq., $f = \frac{1}{T_p}$ [Hz], [Cycle/Sec].

θ → Phase Shift in [°].

• X(t) → always Periodic $X(t) = X(t \pm T_p)$

* Discrete-Time SinuSoidal Signal *



$X(t) = A\sin(2\pi Ft + \theta)$

Replace each t By (nTs)

f → Cycle/Sec
fs → Sample/Sec
F → Cycle/Sample

#

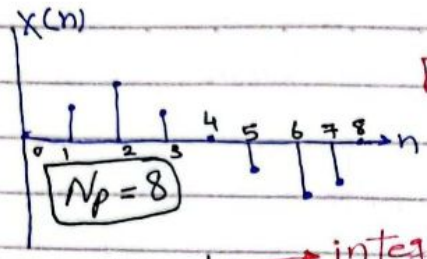
$$X(n) = A\sin(2\pi F nTs + \theta) = A\sin(2\pi \frac{F}{Fs} n + \theta)$$

→ disc-Time SinuSoidal Signal Can Be expressed using:

$$X(n) = A\sin(2\pi F_n + \theta)$$

where: $F = \frac{f}{fs}$ [cycle/sample]

- F: digital Freq.
- fs: Sampling Freq.
- f: Continous-Time Signal



For x(n) To Be Periodic eVery N-SamPle, F must Be Rational number. أي كسر

$$F = \frac{K}{N_P}$$ ← integer num.

N_P ← Fundamental Period (n. of SamPle in one cycle)

in Gobe $x(n) = x(n+N_P)$

#Proof

$$x(n) = A\sin(2\pi Fn + \theta)$$

note: $\theta + 2\pi = \theta$

→ For Periodic Signal

$$x(n) = x(n+N_P)$$

→ GeT $x(n+N_P)$ $(n+N_P)$ بـ f(n) كل

$$x(n+N_P) = A\sin(2\pi F(n+N_P) + \theta)$$

$$x(n+N_P) = A\sin(2\pi Fn + 2\pi F N_P + \theta)$$

if $FN_P = K$, $K = 1,2,3,4,---$

$$\therefore x(n+N_P) = A\sin(2\pi Fn + \theta) = x(n)$$

→ The general Condition Probability 2

$$F = \frac{K}{N_P}$$

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Example 1: Check the Periodicity of the Following:

$$X(t) = 3\sin(300t + 30^\circ)$$

Continuous

$$X(t) = 3\sin(300t + 30^\circ)$$

$$x(t) = A\sin(2\pi f t + \theta)$$

$2\pi f = 300 \rightarrow f = \frac{150}{\pi}$ Hz

$\therefore T_p = \frac{1}{f} = \frac{\pi}{150}$ sec

Note:
$\theta + 2\pi = \theta$

$$x(t+T_p) = 3\sin(300(t+T_p) + 30^\circ)$$
$$= 3\sin(300(t + \tfrac{\pi}{150}) + 30^\circ)$$
$$= 3\sin(300t + 2\pi + 30^\circ)$$
$$= 3\sin(300t + 30^\circ)$$

$\therefore x(t) = x(t+T_p)$ and Signal Periodic every $\frac{\pi}{150}$ sec.

Example 2: Check the Periodicity of the Following:

$$X(n) = 3\sin(\tfrac{30}{105}\pi n + 30^\circ)$$

Discrete

$$X(n) = 3\sin(\tfrac{30}{105}\pi n + 30^\circ)$$

$$x(n) = A\sin(2\pi F n + \theta)$$

$F = \frac{30}{105 \times 2}$ cycle / Sample

$\therefore F = \frac{1}{7} = \frac{k}{N_p}$

$$X(n+N_p) = 3\sin(\tfrac{30}{105}\pi(n+N_p) + 30^\circ)$$
$$= 3\sin(\tfrac{30}{105}\pi(n+7) + 30^\circ)$$
$$= 3\sin(\tfrac{30}{105}\pi n + 2\pi + 30^\circ)$$
$$= 3\sin(\tfrac{30}{105}\pi n + 30^\circ)$$

$x(n) = x(n+N_p)$
↳ Signal Periodic every 7 Samples

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Example 3: check the Periodicity of the following:

$$X(n) = 3\sin\left(\frac{30}{105}n + 30^\circ\right)$$

$$X(n) = A\sin(2\pi F n + \theta)$$

(Irrational) $2\pi F = \frac{30}{105} \rightarrow F = \frac{30}{210\pi}$ cycle/Sample

The Signal is APeriodic (non-Periodic) Signal.

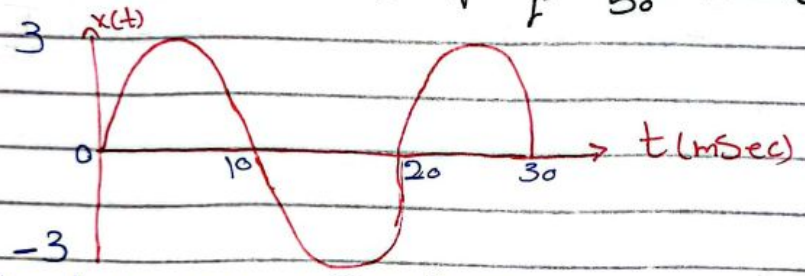
Example 4: $X(t) = 3\sin(100\pi t)$

(a) Plot Signal in the Time interval [0, 30 mSec].

$X(t) = 3\sin(100\pi t)$ (Continous)

$X(t) = A\sin(2\pi f t + \theta)$

$A = 3$, $f = 50$ Hz, $T_p = \frac{1}{f} = \frac{1}{50} = 20$mSec, $\theta = 0$



(b) check the Signal Periodicity

$$X(t+T_p) = 3\sin(100\pi(t+T_p))$$
$$= 3\sin(100\pi t + 100\pi T_p)$$
$$= 3\sin(100\pi t + 2\pi)$$
$$= 3\sin(100\pi t)$$

The Signal is Periodic every 20mSec.

(c) Using $F_s = 300$ Sample/Sec, Can we ReConstruct original Signal Successfully from its Samples, IF yes, Find Fundamental Periodic?

Nyquist $\Rightarrow F_s \geq 2F_{max} \rightarrow 50$ Hz $\quad 300 \geq 2 \times 50$ ✓

they we Can Construct original Signal from Samples.

AT $F_s = 300$ Sample/Sec $\Rightarrow F = \frac{f}{f_s} = \frac{50}{300} = \frac{1}{6}$ cycle/Sample.

$F = \frac{K}{N_p} = \frac{1}{6}$ $\therefore N_p = 6$ Symbols

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② Check the Signal Periodicity

- Get $x(n)$ $t \rightarrow nT_s$

$$X(t) = 3\sin(100\pi t)$$

$$x(n) = 3\sin(100\pi n T_s) \qquad T_s = \frac{1}{F_s} = \frac{1}{300}\ \text{sec}$$

$$\therefore x(n) = 3\sin\left(\frac{100}{300}\pi n\right) = 3\sin\left(\frac{\pi n}{3}\right)$$

$$x(n) = A\sin(2\pi n F + \theta)$$

$$2F = \frac{1}{3} \rightarrow F = \frac{1}{6}\ \text{cycle / sample}$$

$$F = \frac{K}{N_p} = \frac{1}{6} \rightarrow \{N_p = 6\}\ \text{samples.}$$

$$x(n+N_p) = 3\sin\left(\frac{\pi(n+N_p)}{3}\right)$$

$$= 3\sin\left(\frac{\pi n}{3} + \frac{\pi N_p}{3}\right)$$

$$= 3\sin\left(\frac{\pi n}{3} + 2\pi\right) = 3\sin\left(\frac{\pi n}{3}\right)$$

$\therefore$ Signal is Periodic every 6 samples.

③ Compute samples values during one cycle of $x(n)$ and plot $x(n)$ and $x(t)$ on the same figure.

$N = 6 \quad n = (0, 1, 2, 3, 4, 5)$

$x(n) = 3\sin\left(\frac{\pi n}{3}\right)$

$x(0) = 0$

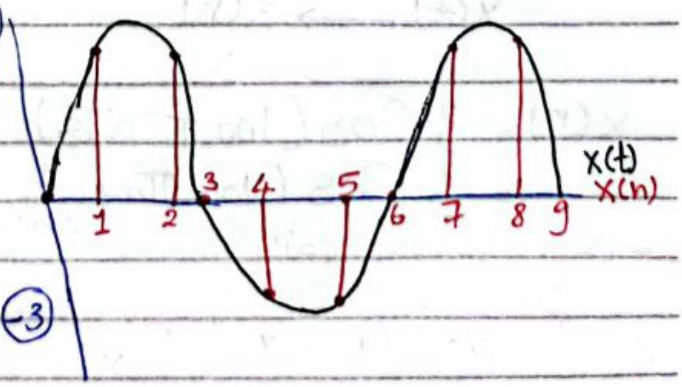
$x(1) = 2.6$

$x(2) = 2.6$

$x(3) = 0$

$x(4) = -2.6$

$x(5) = -2.6$



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(E) Can you find a sampling rate Fs, such that the signal reaches its peak value?

$x(n) = 3 \Rightarrow$ Peak

$x(n) = 3 \sin\left(\frac{100\pi n}{Fs}\right) = 3$

$\sin\left(\frac{100\pi n}{Fs}\right) = 1$

$\frac{100\pi n}{Fs} = \sin^{-1}(1) = 90 = \frac{\pi}{2}$

$Fs = 200n \quad (n = 1, 2, 3, \cdots)$

Example 5: $x(t) = 3\cos(100\pi t)$

(a) using twice minimum sampling freq., obtain the sample values during one cycle of $x(n)$?

$x(t) = 3\cos(100\pi t)$

$x(t) = A\cos(2\pi f t + \theta)$

$f \rightarrow 50\,Hz, \quad T_p = 20\,mSec$

$Fs \geq 2F_{max}$

$Fs \geq 2 \times 50$

$\rightarrow$ Nyquist $Fs \geq 100\,Hz$

$F_{Nyquist} = Fs|_{min} = 100\,Hz$

using twice $Fs|_{min} \rightarrow (Fs = 200\,Hz)$

$x(t) \longrightarrow x(n)$

$t \rightarrow nTs \rightarrow \frac{1}{Fs} = \frac{1}{200} = 5\,mSec$

$x(n) = 3\cos(100\pi nTs)$

$= 3\cos\left(100\pi n \times \frac{1}{200}\right) = 3\cos\left(\frac{n\pi}{2}\right)$

↳ calc. sample value

$F = \frac{f}{fs} = \frac{k}{Np} = \frac{1}{4} \rightarrow Np = 4$ samples

$x(0) = 3 \quad x(1) = 0$

$x(2) = -3 \quad x(3) = 0$

Chapter 2

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Discrete Fourier Transform (discrete & Periodic)

1. discrete Fourier Transform (DFT)
2. Fast Fourier Transform (FFT)

$e^{\pm j\theta} \rightarrow (\cos\theta \pm j\sin\theta)$

discrete Fourier Transform (DFT)

$$\underbrace{X(k)}_{\text{Spectrum domain}} = \sum_{n=0}^{N-1} \underbrace{x(n)}_{\text{Time domain}} e^{-j\frac{2\pi}{N}nk}$$

n → Sample number [0,1,2,3]
N → n.of Sample [N=4]
k → harmonic number [0,1,2,3]

Example: Find the Frequency Component X(k) of Sequences:

(1) $x(n) = \{-2, 1, 0.5, 3\}$ (N=4)
$x(0)\ x(1)\ x(2)\ x(3)$

$$X(k) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{\pi}{2}nk}$$

(Sum.) يجب كتابة عناصر أعلى الـ
n=0:3 وتبدأ من الصفر حتى

$$X(k) = x(0) + x(1)e^{-j\frac{\pi}{2}k} + x(2)e^{-j\pi k} + x(3)e^{-j\frac{3\pi}{2}k}$$

$$X(k) = -2 + 1e^{-j\frac{\pi}{2}k} + 0.5\,e^{-j\pi k} + 3e^{-j\frac{3\pi}{2}k} \quad (k=0,1,2,3)$$

$X(0) = -2 + 1 + 0.5 + 3 = \boxed{2.5}$ AT(k=0)

$X(1) = -2 + e^{-j\frac{\pi}{2}} + 0.5e^{-j\pi} + 3e^{-j\frac{3\pi}{2}}$ AT(k=1)

$= -2 + \left[\cos(\tfrac{\pi}{2}) - j\sin(\tfrac{\pi}{2})\right] + 0.5\left[\cos(\pi) - j\sin(\pi)\right] + 3\left[\cos\tfrac{3\pi}{2} - j\sin\tfrac{3\pi}{2}\right]$
(cos(π/2) → zero, sin(π/2) → 1; cos(π) → -1, sin(π) → 0; cos(3π/2) → 0, sin(3π/2) → -1)

$= -2 - j - 0.5 + 3j = \boxed{-2.5 + 2j}$

$X(2) = -2 + e^{-j\pi} + 0.5e^{-2j\pi} + 3e^{-j3\pi} = -2 - 1 + 0.5 - 3 = \boxed{-5.5}$

$X(3) = -2 + e^{-j\frac{3\pi}{2}} + 0.5e^{-j3\pi} + 3e^{-j\frac{9\pi}{2}} = -2 + j - 0.5 - 3j$

$= \boxed{-2.5 - 2j}$

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② Plot Amplitude & Phase Spectrum?

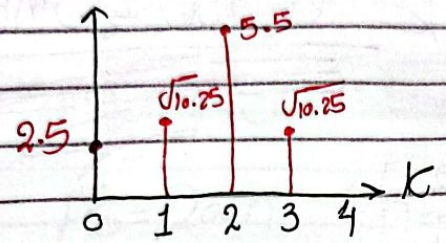
$$X(k) = \{2.5, -2.5+2j, -5.5, -2.5-2j\}$$

① the Amplitude Spectrum $|X(k)|$ *(Positive Always)

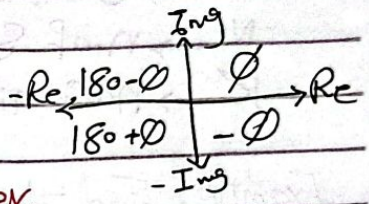
$$|X(k)| = \sqrt{Re^2 + Img^2}$$

$$|X(k)| = \{2.5, \sqrt{2.5^2+2^2}, 5.5, \sqrt{2.5^2+2^2}\}$$

$$= \{2.5, \sqrt{10.25}, 5.5, \sqrt{10.25}\}$$



② The Phase Spectrum $\varphi(k)$



Real only: $\varphi(k) = \begin{cases} 0 & +ve \\ 180^\circ & -ve \end{cases}$

Img only: $\varphi(k) = \begin{cases} 90^\circ & +ve \\ 270^\circ & -ve \end{cases}$

Complex: $\varphi(k) = \tan^{-1}\left(\frac{Img}{Real}\right)$

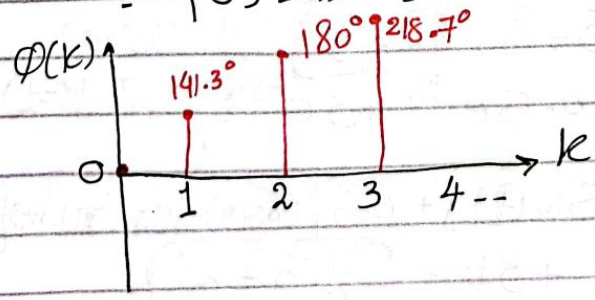
$$X(k) = \{2.5, -2.5+2j, -5.5, -2.5-2j\}$$

$$\varphi(k) = \left\{0, \tan^{-1}\left(\frac{2}{2.5}\right), 180, \tan^{-1}\left(\frac{2}{2.5}\right)\right\}$$

لاحظ كده بجرد الربع اللي شغال فيه

$$\varphi(k) = \{0, 180-\tan^{-1}(\tfrac{2}{2.5}), 180^\circ, 180+\tan^{-1}(\tfrac{2}{2.5})\}$$

$$= \{0, 141.3^\circ, 180^\circ, 218.7^\circ\}$$



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③ Check the Parseval's theory?

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

$x(n) = \{-2, 1, 0.5, 3\}$ $(N=4)$

$X(k) = \{2.5, -2.5+2j, -5.5, -2.5-2j\}$

$|X(k)| = \{2.5, \sqrt{10.25}, 5.5, \sqrt{10.25}\}$

$L.H.S = (-2)^2 + (1)^2 + (0.5)^2 + (3)^2 = 14.25$

$$R.H.S = \frac{1}{4}\left[(2.5)^2 + (\sqrt{10.25})^2 + (5.5)^2 + (\sqrt{10.25})^2\right]$$

$= 14.25$

∴ Parseval's is Verified

To get inverse DFT (IDFT) of $X(k)$?

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{j\frac{2\pi}{N}nk}$$

$X(k) = \{2.5, -2.5+2j, -5.5, -2.5-2j\}$

$$x(n) = \frac{1}{4} \sum_{k=0}^{3} X(k) e^{j\frac{\pi}{2}nk}$$

$$x(n) = \frac{1}{4}\left[X(0) + X(1)e^{j\frac{\pi n}{2}} + X(2)e^{j\pi n} + X(3)e^{j\frac{3\pi n}{2}}\right]$$

$$= \frac{1}{4}\left[2.5 + (-2.5+2j)e^{j\frac{\pi n}{2}} + (-5.5)e^{j\pi n} + (-2.5-2j)e^{j\frac{3\pi n}{2}}\right]$$

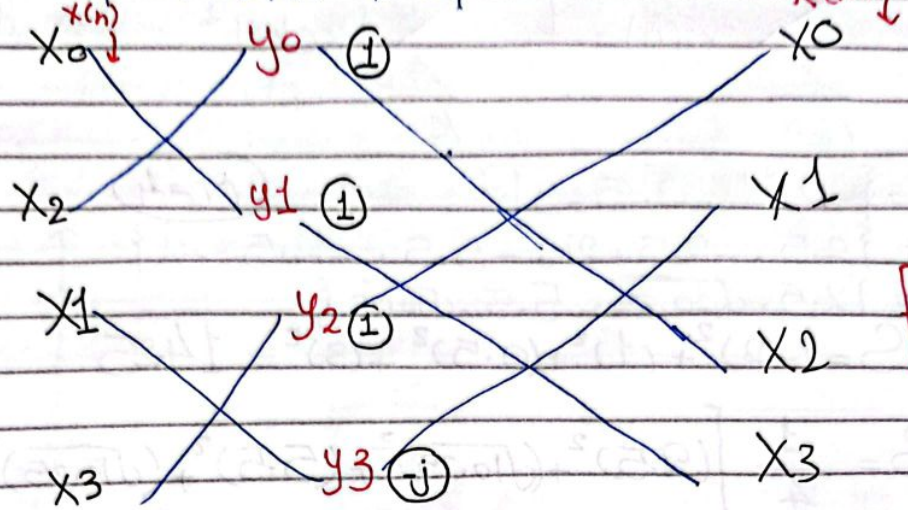
(put $n = 0, 1, 2, 3$)

$X(0) = -2 \quad X(1) = 1 \quad X(2) = 0.5 \quad X(3) = 3$

$e^{\pm j\theta} \rightarrow \cos(\theta) \pm j\sin(\theta)$

Fast Fourier Transform (FFT)

$x(n) = \{x_0, x_1, x_2, x_3\}$



$y_0 = x_0 + x_2$ $\quad X(0) = y_0 + y_2$

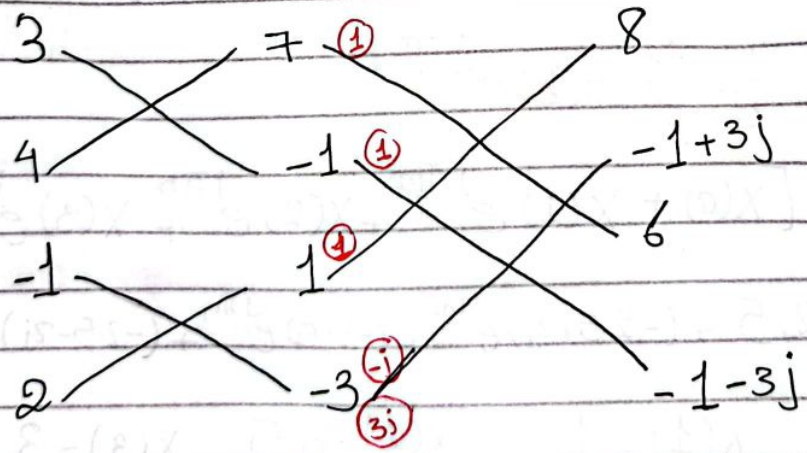
$y_1 = x_0 - x_2$ $\quad X(1) = y_1 - (j)\, y_3$

$y_2 = x_1 + x_3$ $\quad X(2) = y_0 - y_2$

$y_3 = x_1 - x_3$ $\quad X(3) = y_1 + (j)\, y_3$

* Find the Freq Component $X(k)$ of the following:

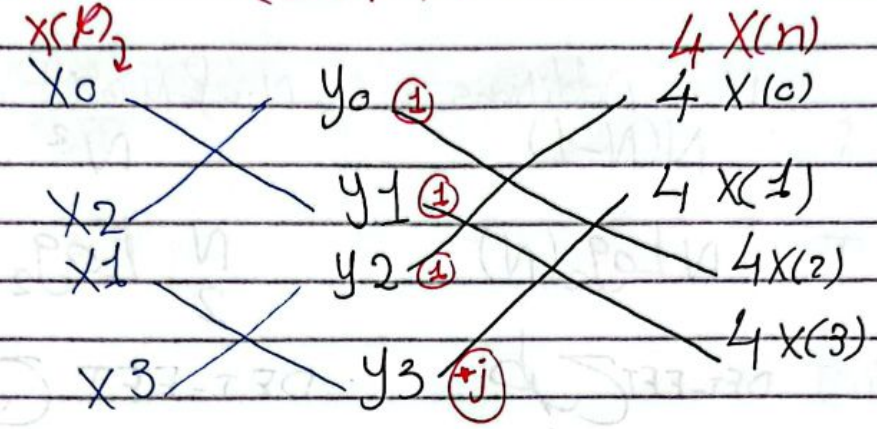
$x(n) = \{3, -1, 4, 2\}$ (x_0, x_1, x_2, x_3)



$X(k) = \{8, -1+3j, 6, -1-3j\}$

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Inverse Fast Fourier Transform (IFFT)



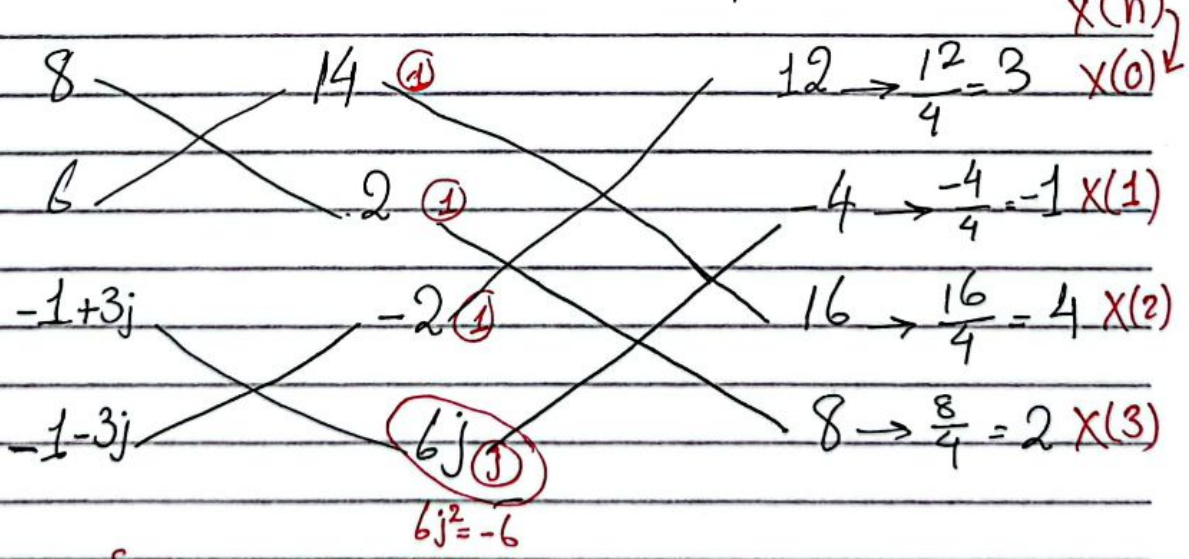
1	1
1	j

$y_0 = X_0 + X_2$
$y_1 = X_0 - X_2$
$y_2 = X_1 + X_3$
$y_3 = X_1 - X_3$

$4X(0) = y_0 + y_2$
$4X(1) = y_1 + (j)y_3$
$4X(2) = y_0 - y_2$
$4X(3) = y_1 - (j)y_3$

#Continay Lab Example:

$x(k) = \{8, -1+3j, 6, -1-3j\}$



$x(n) = \{3, -1, 4, 2\}$

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Comparison Between DFT & FFT?

	N. of Additions	N. of Multiplications
DFT	$N(N-1)$	N^2
FFT	$N\log_2(N)$	$\frac{N}{2}\log_2(N)$
Saving	DFT-FFT	DFT-FFT

Chapter 3

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Correlation & Convolution

#Correlation#

↳ Measure degree of Similarity Between 2 Signals.

There Are 2 Types:

1. Cross-Correlation (r_{xy})
   ↳ Measured for 2 different Signals.
2. Auto-Correlation (r_{xx})
   ↳ Measured for the Same Signal.

Cross-Correlation

$$r_{xy} = \frac{1}{N_{new}} \sum_{n=0}^{N-1} x(n)\, y(n+j)$$

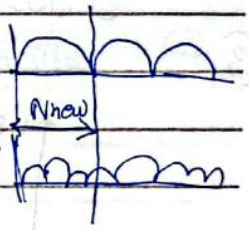
N_{new} ← No. of Samples

j ← Lag (Shift) Value

2 Types of Cross-Correlation

↳ 1. Linear $N_{new} = N_1 + N_2 - 1$

↳ 2. Cyclic $N_{new} = MAX(N_1, N_2)$



Ex: (1) Calc. Linear Cross-correlation Between

$X_1 = \{1, 2, -3, 4\}$, $X_2 = \{3, -2, 2\}$

(N_1 = 4, N_2 = 3)

$N_{new} = N_1 + N_2 - 1 = 4 + 3 - 1 = 6$ Samples

$X_1(n)$	1	2	-3	4	0	0	j	$r_{12}(j)$
	(3)	-2	2	0	0	0	0	$-7/6 = -1.167$
	(-2)	2	0	0	0	(3)	1	$2/6 = 0.333$
$X_2(n+j)$	(2)	0	0	0	3	-2	2	$2/6 = 0.333$
	(0)	0	0	3	-2	2	3	$12/6$
	(0)	0	3	-2	2	0	4	$-17/6$
	(0)	3	-2	2	0	0	5	$20/6$
	3	-2	2	0	0	0	6	$-7/6$

$j \rightarrow 0 : N_{new} - 1$

X_1, X_2 يضرب

(N) نجمع ونقسم على

Shift Left and Rotate

(N) نجمع ونقسم على N_{new} (بعد ضرب r_{12})

هيك جبنا أول حالة ثاني



② Calc. Cross-correlation Coefficient $\rho_{12}(j)$:

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}} \sqrt{\sum_{n=0}^{N-1} x_1^2(n) \cdot \sum_{n=0}^{N} x_2^2(n)}} = \frac{r_{12}(j)}{3.7639}$$

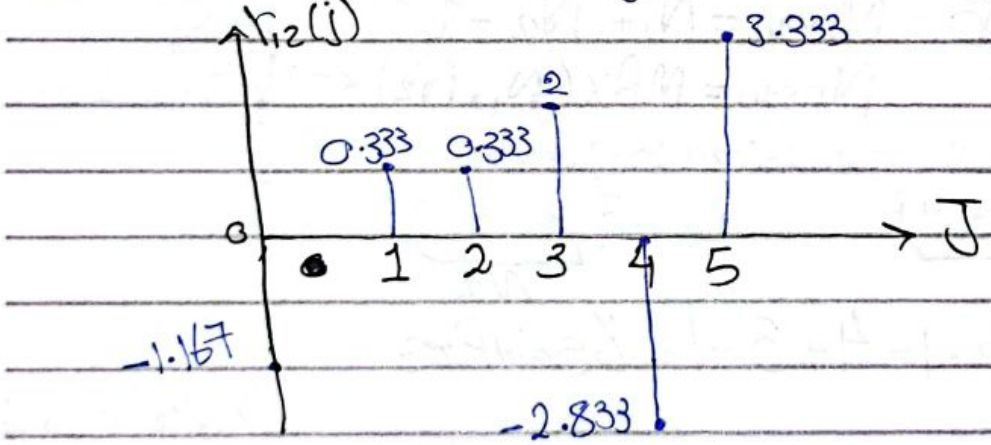
N_{new} ⑥

$x_1^2(n)$: $\{1, 2, -3, 4\}$ → $1^2 + 2^2 + (-3)^2 + (4)^2 = 30$

$x_2^2(n)$: $\{3, -2, 2\}$ → $3^2 + (-2)^2 + (2)^2 = 17$

$r_{12}(j)$	$\rho_{12}(j)$
-1.167	$\frac{-1.167}{3.7639} = -0.31$
0.333	$\frac{0.333}{3.7639} = 0.0886$
0.333	// $= 0.0886$
2	$\frac{2}{3.7639} = 0.6314$
-2.833	$\frac{-2.833}{3.7639} = -0.7528$
3.333	$\frac{3.333}{3.7639} = 0.8856$

③ Plot correlation function against the Lag number:



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5) Cyclic Cross-Correlation

Ex: ① Calc. The Cyclic Cross-Correlation Between

$X_1 = \{1, 2, -3, 4\}$ and $X_2 = \{3, -2, 2, -1\}$

$N_1 = 4$ $\quad N_2 = 4$

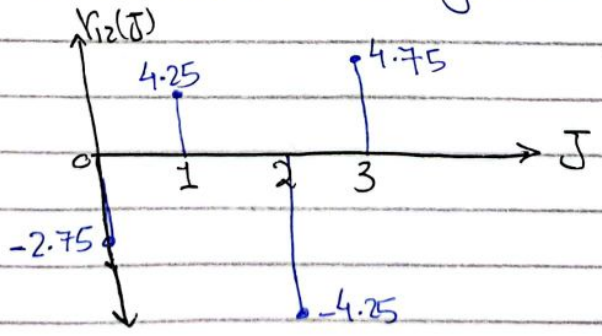
$\therefore N_{new} = Max(N_1, N_2) = 4$

$X_1(n)$	1	2	-3	4	J	$r_{12}(J)$	$\rho_{12}(j)$
	③	-2	2	-1	0	$-11/4 = -2.75$	-0.473
$X_2(n+j)$	-2	2	-1	③	1	$17/4 = 4.25$	0.7315
	2	-1	3	-2	2	$-17/4 = -4.25$	-0.7315
	-1	3	-2	2	3	$19/4 = 4.75$	0.8176

② Calc. cross-Correlation Coefficient $\rho_{12}(j)$

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}} \sqrt{\sum_{n=0}^{N-1} x_1^2(n) \cdot \sum_{n=0}^{N-1} x_2^2(n)}} = \frac{r_{12}(j)}{\frac{1}{4}\sqrt{(30 \times 18)}} = \frac{r_{12}(j)}{5.8095}$$

③ Plot the Correlation function against Lag number:



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Auto-Correlation

• Calc. the auto correlation function $r_{11}(j)$ For the $X_1 = \{1, 2, -3, 4\}$

Shift Left without Rotation

$$r_{11}(j) = \frac{1}{N_{new}} \sum X_1(n) \cdot X_1(n+j)$$

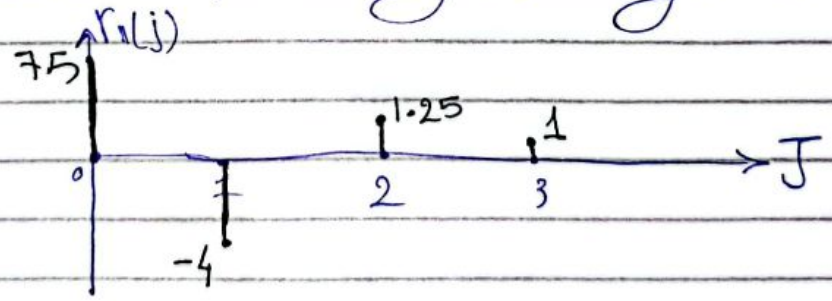
$N_{new} = N_1 = 4$ Samples

$X_1(n)$	1	2	-3	4	J	$r_{11}(J)$		$\rho_{11}(j)$
	1	2	-3	4	0	$30/4 = 7.5$		1
$X_1(n+j)$	2	-3	4	0	1	$-16/4 = -4$		-0.5333
	-3	4	0	0	2	[illegible]	$5/4 = 1.25$	0.16667
	4	0	0	0	3	$4/4 = 1$		0.1333

• Calc. auto Correlation Coefficient $\rho_{11}(j)$:

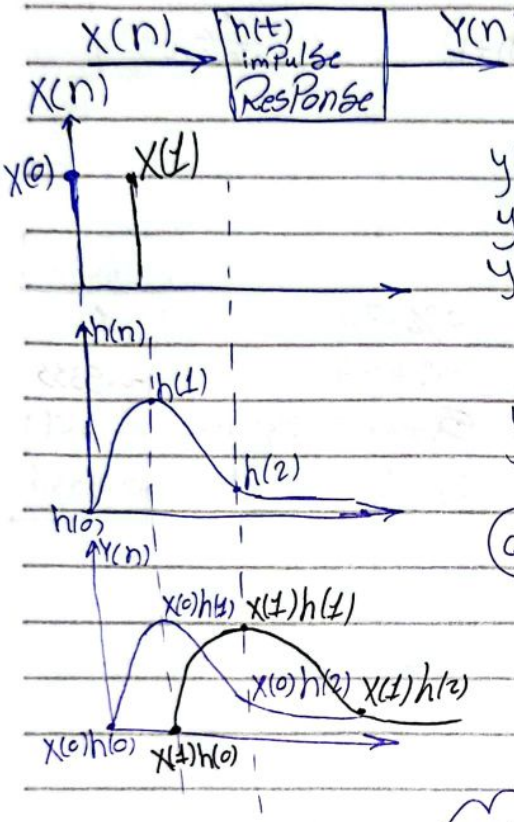
$$\rho_{11}(j) = \frac{r_{11}(j)}{\frac{1}{N_1} \sum_{n=0}^{N_{new}-1} X_1^2(n)} = \frac{r_{11}(j)}{7.5}$$

• Plot Correlation Function against Lag number:



Convolution operation

↳ how the input interacts with System To Produce output.



$Y(n) = X(n) * h(n)$

$y(0) = X(0)h(0)$

$y(1) = X(0)h(1) + X(1)h(0)$

$y(2) = X(0)h(2) + X(1)h(1) + X(2)h(0)$

$$y(n) = \sum_{m=0}^{n} h(m)\,X(n-m)$$

or

$$y(n) = \sum_{m=0}^{n} X(m)\,h(n-m)$$

بحيث يعتمد على حالتين.

[1] Inversion. [2] Shift.

#Linear Convolution (Shift Right)

$X(m) = \{0,0,0,1,-2,3\}$

Ex: Calc. The System out Put when the in Put is $X(n) = \{1,-2,3\}$; if The System im Pulse Response is given By $h(n) = \{-2,2,1,4\}$.

$h(-m) = \{4,1,2,-2,0,0\}$

$N_{new} = N_1 + N_2 - 1 = 3 + 4 - 1 = 6$

X(m)	0	0	0	1	-2	3	n	y(n) (o/p)
h(n-m)	4	1	2	-2	0	0	0	-2
	0	4	1	2	-2	0	1	6
	0	0	4	1	2	-2	2	-9
	0	0	0	4	1	2	3	8
	0	0	0	0	4	1	4	-5
	0	0	0	0	0	4	5	12

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#Circular Convolution

Calc. The System out Put when The input Sequence y
$x(n)=\{1,-2,3,-1\}$ if the System impulse given By $h(n)=\{-2,2,1,4\}$.

$x(m)=\{1,-2,3,-1\}$

$h(-m)=\{-2,4,1,2\}$ أول عنصر ثابت والباقي بعكس

$x(m)$	1	-2	3	-1	n	$y(n)$
	-2	4	1	2	0	-9
$h(n-m)$	2	-2	4	1	1	17
	1	2	-2	4	2	-13
	4	1	2	-2	3	10

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CH4 Z-Transform

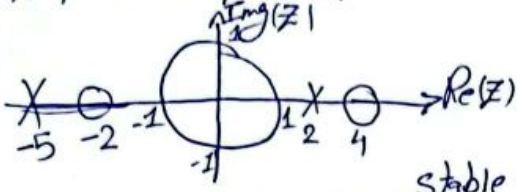
$H(Z)=\frac{1-2Z^{-1}-8Z^{-2}}{1+3Z^{-1}-10Z^{-2}}$

[1] Plot the Pole-Zero diagram

$H(Z)=\frac{Z^2-2Z-8}{Z^2+3Z-10}=\frac{(Z-4)(Z+2)}{(Z-2)(Z+5)}$

O Zeros → {4, -2}

X Poles → {2, -5}



[2] check the Stability (Poles)

stable / critically stable / unstable

*System is unstable Because at Least one Poles outside The unit circule.

[3] Find ROC

↳ All value that make H(Z) is Finite

*ROC is entire value of Z-Plane expect Z = {2, -5}

[4] Find System difference eq.

$H(Z)=\frac{1-2Z^{-1}-8Z^{-2}}{1+3Z^{-1}-10Z^{-2}}=\frac{Y(Z)}{X(Z)}$

$X(Z)-2X(Z)Z^{-1}-8X(Z)Z^{-2}=Y(Z)+3Y(Z)Z^{-1}-10Y(Z)Z^{-2}$

↓

$X(n)-2X(n-1)-8X(n-2)=y(n)+3y(n-1)-10y(n-2)$

$\therefore y(n)=X(n)-2x(n-1)-8x(n-2)-3y(n-1)+10y(n-2)$

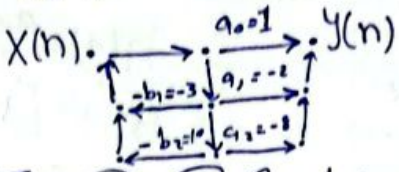
IZT → Power Series → Long divisision / Recusively
IZT → Partial fraction
IZT → Residue

[5] Plot the System Signal Flow graph

$a_0 \to 1$

$a_1 \to -2$ $b_1 \to 3$

$a_2 \to -8$ $b_2 \to -10$



[6] Find im Pulse Response h(n) using The Power Series method (Recusively)

$a_0 \to 1$ $b_0 \to 1$

$a_1 \to -2$ $b_1 \to 3$

$a_2 \to -8$ $b_2 \to -10$

$h(0)=a_0/b_0=1$ #

$h(n)=[a_n-\sum_{i=1}^{n} h(n-i)b_i]/b_0$

$h(1)=[a_1-h(0)b_1]/b_0=-5$ #

$h(2)=[a_2-h(1)b_1-h(0)b_2]/b_0=17$ #

$h(3)=[a_3-h(2)b_1-h(1)b_2-h(0)b_3]/b_0=-101$ #

[7] Find im Pulse Response using Residue

$H(Z)=\frac{Z^2-2Z-8}{Z^2+3Z-10}$

$F(Z)=Z^{n-1}H(Z)=\frac{Z^{n-1}(Z-4)(Z+2)}{(Z-2)(Z+5)}$

has 2 Poles at $Z=\{-5, 2\}$

$h(n)=Res[F(Z),-5]+Res[F(Z),2]$

$Res[F(Z),-5]=(Z+5)F(Z)=\frac{Z^{n-1}(Z-4)(Z+2)}{(Z-2)(Z+5)}\Big|_{Z=-5}$

$=\frac{(-5)^{n-1}(-9)(-3)}{(-7)}=(-5)^{n-1}\frac{27}{-7}$

$Res[F(Z),2]=(Z-2)F(Z)$

$=\frac{Z^{n-1}(Z-4)(Z+2)}{(Z+5)}\Big|_{Z=2}$

$=\frac{2^{n-1}(-2)(4)}{7}=\frac{-6}{7}(2)^{n-1}$

#IZT

$$X(Z)=\frac{1+2Z^{-1}+Z^{-2}}{1-Z^{-1}+0.3561Z^{-2}}$$

#Long division

$$1-Z^{-1}+0.3561Z^{-2}\;\overline{)\;1+2Z^{-1}+Z^{-2}}\;\;1+3Z^{-1}+3.6439Z^{-2}+2.57Z^{-3}$$

$$\ominus\; 1 \oplus Z^{-1} \ominus 0.3561Z^{-2}$$

$$3Z^{-1}+0.6439Z^{-2}$$

$$\ominus\; 3Z^{-1} \oplus 3Z^{-2} \ominus 1.0683Z^{-3}$$

$$3.6439Z^{-2}-1.0683Z^{-3}$$

$$\ominus\; 3.6439Z^{-2} \oplus 3.6439Z^{-3} \ominus 1.297Z^{-4}$$

$$2.57Z^{-3}-1.297Z^{-4}$$

$$\therefore X(n)=\{1,3,3.6439,2.57\}$$

#Recusively

$a_0 \to 1$ $\quad b_0 \to 1$

$a_1 \to 2$ $\quad b_1 \to -1$

$a_2 \to 1$ $\quad b_2 \to 0.3561$

$h(0)=a_0/b_0=1$

$h(n)=\left[a_n-\sum_{i=1}^{n} h(n-i)b_i\right]/b_0$

$h(1)=[a_1-h(0)b_1]/b_0=3$

$h(2)=[a_2-h(1)b_1-h(0)b_2]/b_0=3.6439$

$h(3)=[a_3-h(2)b_1-h(1)b_2-h(0)b_3]/b_0$

($a_3 \to 0$, $h(0)b_3 \to 0$)

$=2.57$

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CH5 Digital Filter

FIR

- $H(Z) = h(0) + h(1)Z^{-1} + h(2)Z^{-2}$
  $= a_0 + a_1 Z^{-1} + a_2 Z^{-2}$
- Relaization Structure (Block diagram) with Direct method + Lattic.

IIR

- $H(Z) = \frac{a_0 + a_1 Z^{-1} + a_2 Z^{-2}}{1 + b_1 Z^{-1} + b_2 Z^{-2}}$
- Relization Structure (Block diagram) with direct method only.

*For FIR

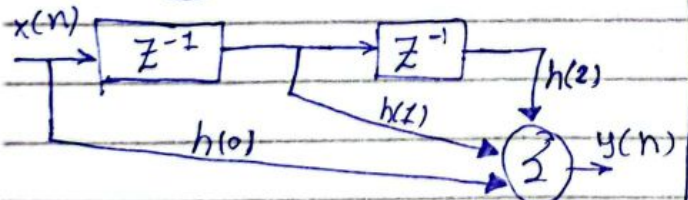
difference eq. From Transfer function:

$$H(Z) = \frac{Y(Z)}{X(Z)} = h(0) + h(1)Z^{-1} + h(2)Z^{-2}$$

$$Y(Z) = h(0)X(Z) + h(1)Z^{-1}X(Z) + h(2)Z^{-2}X(Z)$$

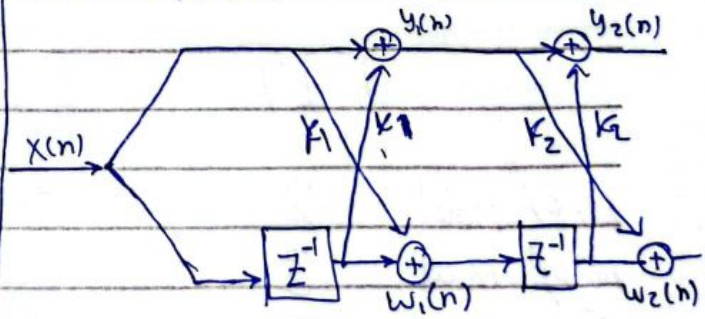
$$y(n) = h(0)X(n) + h(1)X(n-1) + h(2)X(n-2)$$

Block diagram (Direct method)



* No. of Add = Input to Sum − 1 = 3 − 1 = 2
* No. of Mul = 3
* ~~No. of Storage = Input + delay + Coeff~~
  $= 2 + 2 + 3 = 7$
  (2 → Add, 2 → I/P / O/P, 3 → Mul)

Lattic method



Given: $h(1) = 3$ $h(2) = 5$

Find K_1, K_2

$y_1(n) = x(n) + K_1 x(n-1)$

$w_1(n) = K_1 x(n) + x(n-1)$

$y_2(n) = y_1(n) + w_1(n-1)K_2$

$w_2(n) = y_1(n)K_2 + w_1(n-1)$

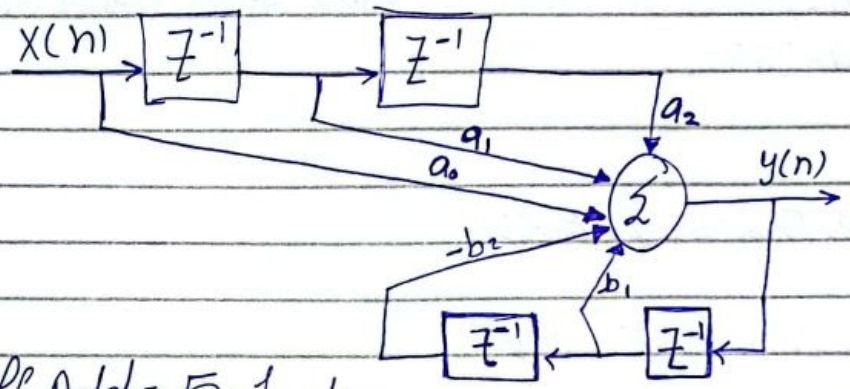
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For IIR :

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{Y(z)}{X(z)}$$

$$Y(z)\left[1 + b_1 z^{-1} + b_2 z^{-2}\right] = X(z)\left[a_0 + a_1 z^{-1} + a_2 z^{-2}\right]$$

$$y(n) = a_0 x(n) + a_1 x(n-1) + a_2 x(n-2) - b_1 y(n-1) - b_2 y(n-2)$$



N.of Add = 5-1 = 4
N.of Mul = 5
Storage = 4+5+2 = 11

Compare Between FIR & IIR

	FIR	IIR
N. of Add	2	4
N. of Mul	3	5
Storage	7	11

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